

Vehicle-In-the-Loop Test Environment for Autonomous Driving with Microscopic Traffic Simulation

1st Tamás Tettamanti

*Dept. of Control for Transportation and Vehicle Systems
Budapest University of Technology and Economics
Budapest, Hungary
tettamanti@mail.bme.hu*

2nd Mátyás Szalai

*Dept. of Control for Transportation and Vehicle Systems
Budapest University of Technology and Economics
Budapest, Hungary
sm1248@hszk.bme.hu*

3rd Sándor Vass

*Dept. of Automotive Technologies
Budapest University of Technology and Economics
Budapest, Hungary
sander.vass@gjt.bme.hu*

4th Viktor Tihanyi

*Dept. of Automotive Technologies
Budapest University of Technology and Economics
Budapest, Hungary
viktor.tihanyi@gjt.bme.hu*

Abstract—In our days, vehicle automation is in a continuous evolutionary phase consisting of experiments, testing and validation. Accelerating the development and deployment of autonomous vehicles and infrastructure is a real demand as these technologies have a great potential to improve traffic safety and resolve road transport problems. The Vehicle-In-the-Loop testing is therefore indispensable throughout the development process. As a potential solution to this need, this paper introduces a new approach for test environment capable to simulate realistic traffic around the autonomous test vehicle. The test car therefore can be put into virtual transportation network by applying real-time microscopic traffic simulation, i.e., the method enables safe vehicle testing. The proposed test environment was proved with real world autonomous car.

Index Terms—autonomous vehicle, Vehicle-In-the-Loop test, CAN communication, microscopic traffic simulation, SUMO TraCI

I. INTRODUCTION

Autonomous vehicle technology will basically transform our life in the near future [1]. The general impact of autonomous transport has been examined in several recent papers. The major social impacts have been summarized as the gains due to safer roads, travel time reduction, more personalized services, improvement of energy efficiency and better parking possibilities [2]–[4]. The most important benefits are the safety-related issues as the majority of traffic accidents (app. 95%) are caused by human imperfection. It is estimated that self-driving technology could eliminate 90% of road traffic accidents [5]. Market researches estimate different scenarios for the penetration of autonomous cars [6]. However, even the worst case scenario [7] predicts the wide spread of fully

driverless vehicles by 2045 at the latest. Nevertheless, it is apparent that in the next decades a significant part of road vehicles will be equipped by highly automated functionalities.

Autonomous driving and related intelligent infrastructure developments open immense possibilities for scientific and technological advance. Beyond the liability issues [8], the main condition of practical deployment is, however, the proper and reliable functioning which is guaranteed by continuous testing and validation in engineering practice [9], [10].

In case of traditional vehicles, manufacturers have plethora of experiences, i.e., millions of test kilometers making possible to correct problems. Moreover, human drivers are educated in a similar way and so have identical knowledge on traffic rules. Contrarily, autonomous vehicles just "start learning" the driving and have not yet gathered that amount of experiences. Due to possible errors testing on public roads with traditional traffic is not allowed in most of the countries. Therefore, it is necessary to create test sites and develop systems that enables autonomous vehicle testing in realistic environments with arbitrary traffic situations. As an example for this, the new test track of Hungary is mentioned which is designed to support the automotive industry throughout the whole process from prototype testing to mass production [11]. The Hungarian proving ground will be realized (close to the city of Zalaegerszeg) through an investment of 130 million Euros by 2019. The planned proving ground is especially dedicated to testing connected and autonomous vehicle functionalities according to the SAE levels [12]. The 250 ha area will incorporate vehicle dynamics testing and validation, environment preparation (obstacles, traffic signs, traffic control, other vehicles, vulnerable road users), complex driving and traffic situations, smart city features.

One of the possible future extension of the test track

The project has been supported by the European Union, co-financed by the European Social Fund EFOP-3.6.2-16-2017-00002 as well as by the János Bolyai Research Scholarship of the Hungarian Academy of Sciences.

will be the Vehicle-In-the-Loop (VIL) testing. There exist several traffic simulation software capable to mimic realistic dynamics. This opens the way to insert the self-driving test car into the virtually generated traffic system. The option of adding autonomous vehicle models to the traffic simulator already exists (e.g., PTV Vissim). However, the usual approach today is limited to fully virtual testing, i.e., no real control of autonomous vehicle functions and behavior implemented [13], [14]. This solutions also insert autonomus vehicles into traffic simulators, but not in a VIL level, only fully simulated. Therefore, in this article, we describe a solution that creates an online connection between a moving (real world) test vehicle and a road traffic simulator. In this way, it is possible to provide data directly to the vehicle control system to ensure that the vehicle's perceived environment become the same as in the traffic simulation software. The advantage of our solution is that SUMO traffic simulator is open source and fully programmable. The built-in framework can be connected to the autonomous test vehicle on the CAN network and any other way thanks to programming in Matlab. An additional advantage is that the connection points are available for any graphical display engines, which is a future work of us though. Thus, arbitrary test traffic situations can be simulated, such as other vehicle interactions, vehicle communication, intelligent crossroads, unexpected circumstances caused by pedestrians, reactions in an accident, etc. One of the greatest advantages of this VIL testing method is cost-effectiveness. Obviously, there is no need to build traffic systems, only the test vehicle must have the software and hardware to process the data of virtual traffic. It is also important to emphasize the safety of simulations, since during the experiments the physical safety of the participants or the proper functioning of a real transport system are not endangered.

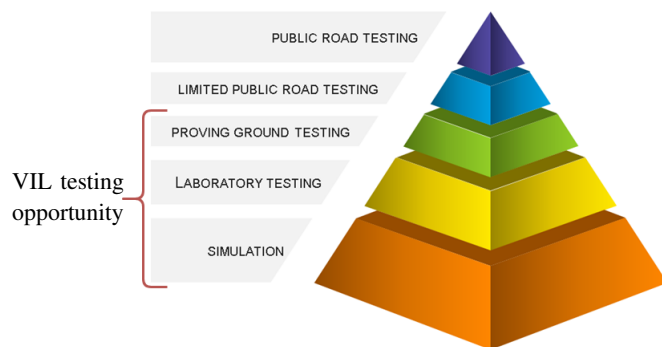


Fig. 1. Autonomous vehicle testing and validation layers [15]

The testing and validation structure for autonomous vehicles can be defined as a 5 layer pyramid depicted by Fig. 1. It is emphasized that the first 3 layers might efficiently use the opportunities of VIL testing, i.e., the phases of Simulation, Laboratory Testing as well as Proving Ground Testing will benefit from the advantages of microscopic traffic simulations.

The paper is structured as follows. In Section II, the concept of VIL testing is introduced. Section III interprets the main elements and the functioning of the VIL testing environment.

In Section IV, the real world test results are discussed. The paper ends up with a short conclusion in Section V.

II. AUTONOMOUS VEHICLE TESTING

Firstly, it is important to distinguish the innovation concerning the vehicle functions and the development of the entire transportation system. In the first case, the focus is on the car itself and its close environment. In the second case, however, one aims to ameliorate the performance of the entire traffic system (vehicle, infrastructure and users). Of course, the automotive industry runs intensive vehicle developments by applying very detailed vehicle model and vehicle dynamics simulations. However, these programs have limited capacity to model traffic situations. From the other side the analysis and design of the whole transport system is performed by the traffic engineering profession, typically carried out with micro- or macroscopic traffic simulation software. These software are already capable of detailed modeling of the whole traffic network, but they operate with simplified vehicle dynamics, e.g., acceleration function can be defined but detailed engine dynamics or powertrain are not included. It is observable that the transition between these two simulation methods is currently limited. It is possible to interface these programs but they are not integrated.

To overcome the above problem, similarly to the "Hardware-In-the-Loop" method, the so-called "Vehicle-In-the-Loop" (VIL) test simulation framework is designed. The essence of this is to create an environment in which a realistic traffic simulation software (with microscopic traffic model) can simulate one or more real (autonomous) test vehicles in real time while other vehicles are implemented as virtual traffic. In this system it is possible to test real world cars together with their autonomous functions and capabilities by displaying real vehicle dynamics in the simulator, while the traffic around is virtually generated. In the VIL test environment, arbitrary circumstances can be realized. One can create vehicles, traffic control objects, pedestrians, accident, or simulate situations with poor visibility where vehicle camera is ineffective and are forced to rely on V2V communication. As shown by Fig. 2, beyond the simulation of the virtual environment around a real vehicle, also communication with virtual systems (other vehicles or the traffic management system) can be realized.

All this makes it possible to carry out cost efficient and safe tests, e.g., on a plain test site without objects.

In order to build a simulation environment, it is necessary to define the main goals. The interface to be created is determined according to the requirements of these simulation targets. The data and control parameters between the test vehicle and the traffic simulation suites are accordingly to be defined. Some of these are trivial, such as the actual vehicle velocity or the vehicle position, but other parameters will depend on the tasks planned to be accomplished. The tasks that can be created in the VIL test environment are classified in the following main categories:

- keeping and changing vehicle speeds according to environmental impact;

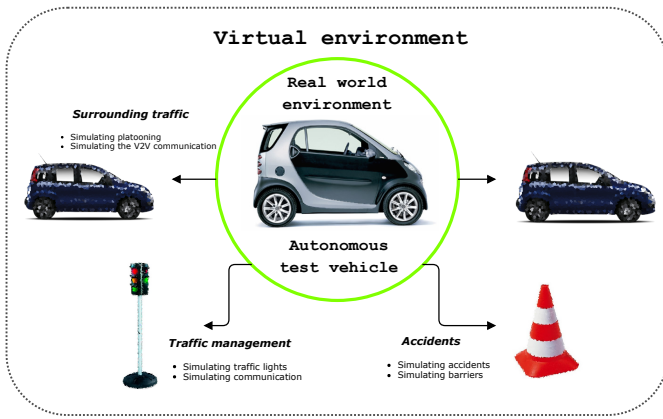


Fig. 2. Simulation possibilities

- test vehicle operation at traffic light;
- emergency brake simulation by virtual interference;
- keeping small headway distance compared to the virtual vehicle ahead;
- platooning situation with two or more real autonomous vehicles.

Of course, each category covers further possible sub-tasks.

III. THE ELEMENTS OF THE VIL TESTING ENVIRONMENT

In the sequel, the main basic building blocks (hardware, software) and the functioning of the whole VIL test environment are introduced.

A. SUMO Microscopic Traffic Simulation Software

Road traffic modeling is commonly used in traffic engineering practice to assist design and validation of newly developed control strategies. Users may choose among many different commercial or open source road traffic simulators. Each of them has advantages or drawbacks depending on the individual demands of the user.

For the purpose of autonomous vehicle tests SUMO (Simulation of Urban MObility) [16] software was chosen due to the flexible programming possibilities. SUMO is a free and open traffic simulation suite for modeling traffic including road vehicles, railway, tram, pedestrians and even ships.

SUMO offers XML based configuration through of which one can design the geometry of any type of road networks and set up simulations in a simple way. However, for several problems this option is not satisfying. This is the case, for example, when the user aims to access and manipulate SUMO objects during the simulation in a dynamic way. To this end, an additional interface called Sumo TraCI (Traffic Control Interface) [17] is offered based on TCP communication which can be programmed in several languages (e.g., Python, Matlab, Java, etc.). SUMO TraCI allows to control and modify a simulation process online. It also enables to retrieve information from any of the elements of the simulation network, such as vehicles, road links, junctions, traffic lights, etc.

The application of microscopic simulation is therefore beneficial for realistic and accurate modeling of the traffic dynamics. Both the simulated traffic network and the autonomous test car become analyzable in this way.

B. Use of CAN Communication

Using CAN based messaging for creating connection between the test vehicle and the simulation software is an obvious solution as it is the most frequently used communication bus networks in automotive industry. In most cases, any parameter related to the vehicle can be found on the CAN bus network. This technology enables communication between the elements of the vehicle, e.g., sensors, ECUs. Thanks to the well-defined CAN network, one can easily write databases that can be interpreted and processed in a Matlab environment by using the standard database of CAN (dbc file). The messages contained in the dbc files can be selected separately easily and handled as a distinct parameter using the Matlab's internal CAN communication module. In this way receiving and transmitting of signals to the network can be realized straightforwardly. In order to be able to read these messages from the CAN bus, the Vector CANCaseXL is applied in our setup. This device with multi-channel communication acts as a node on the network that receives and sends all messages with any identifier. Vector CANCaseXL can be attached to the car's CAN bus easily.

C. Test Vehicle with AutoBox

The Faculty of Transportation Engineering and Vehicle Engineering of BME (Budapest University of Technology and Economics) owns a Smart Fortwo city car. This test car was modified by the university researchers to make it fully capable for "by wire" operation with fail-safe architecture [18]. The system architecture of the Smart is shown by Fig. 3. All execution actuators were modified or at least interrupted using a central Autobox device for realizing the control. (AutoBox is an efficient tool for using dSPACE real-time system to carry out in-vehicle control experiments such as tests for autonomous functionalities.) Throttle had originally a potentiometer based control with analog signal that was interrupted and lead through the device. Steering was solved by additional BLDC motor actuator driven by an external servo drive with torque control fed also from Autobox device. For the gear control the original gear button's parallel interface was substituted by a self developed ECU that can communicate with the central controller via CAN bus. Brake system required high modifications, a linear actuator was built in in order to pull the brake pedal via a bowden. Brake system also includes a spring actuator in case of fail or emergency situations. The vehicle was also equipped with environment sensors (LIDAR and camera) with the capability of lane detection and object detections, however for the VIL tests described in the sequel the LIDAR sensors are not used intentionally. At the same time, LIDAR CAN messages are reproduced by external computer (by Matlab in a simple laptop) to input them for the central controller of the Smart. The vehicle

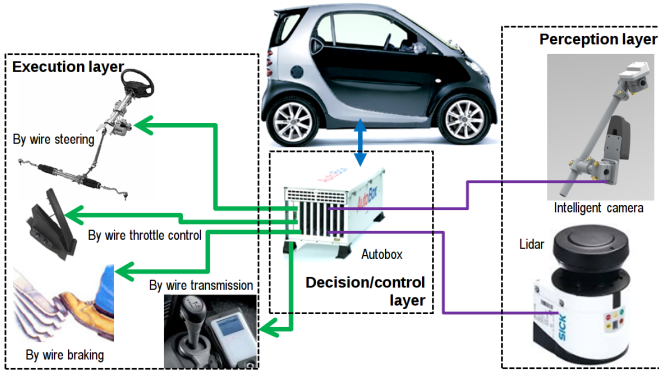


Fig. 3. The system architecture of the Smart for autonomous driving tests

for the time has several automated functions such as lane keeping, lane change and automatic cruise control. Moreover, a high level autonomous function, the traffic jam assist, is also realized combining the existing fundamental driver assistance components.

D. The VIL Test Environment

The whole VIL test environment for autonomous car is depicted by Fig. 4. The environment can be divided into two main parts:

- 1) the test vehicle with own computer logic for autonomous functions and its physical environment (where it moves);
- 2) the computer realizing the VIL test environment (simulation software).

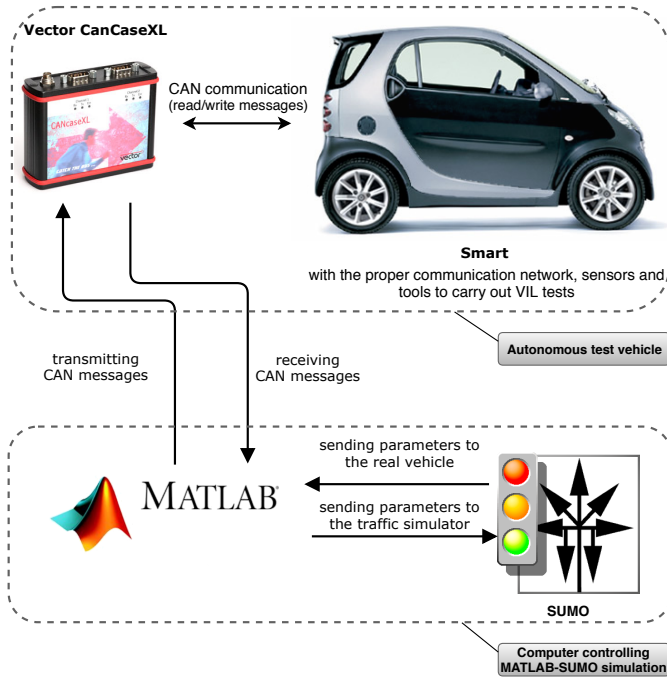


Fig. 4. The whole test environment for autonomous car

The elements of the system devices are connected via CAN communication channels, i.e., Vector CANCaseXL controls

the messages specified for the CAN bus. CAN messages are processed by Matlab and then transferred to the SUMO traffic simulator via the TraCI interface. Thus, a complete communication channel is created between the Smart vehicle (central controller) and the SUMO traffic simulator.

IV. REAL WORLD TEST IN THE DEVELOPED VIL TEST ENVIRONMENT

As a real world demonstration of the developed VIL simulation environment one of the autonomous functions of the Smart, the traffic jam assist, was tested. The test environment is depicted in Fig. 5. The area that serves as a car park in the university campus of BME was modeled in the traffic simulator. The test scenario simulated the motion of two cars on a specific route (Fig. 6), i.e.,

- 1) the virtual car (blue) is going ahead and
- 2) the real autonomous vehicle (green) follows it.

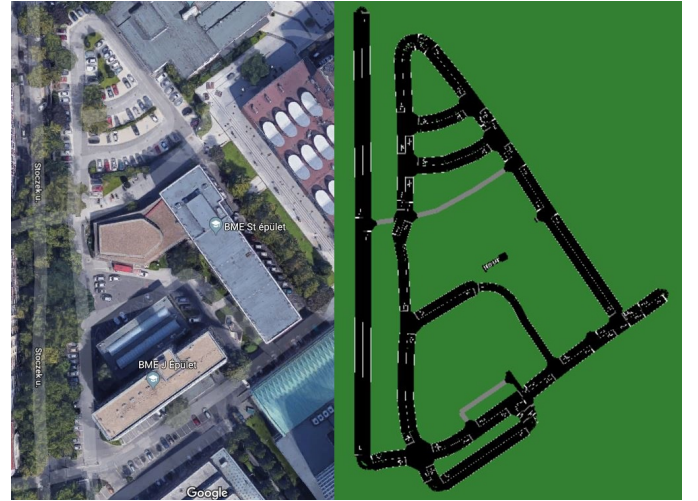


Fig. 5. The location of the test (the campus of Budapest University of Technology and Economics, GPS: 47.478488, 19.056098)

During the experiment, the acceleration and braking maneuvers of the real vehicle was directly fitted into the traffic simulator while the motion of the virtual car was simulated by SUMO.

At the beginning of the test, the two vehicles started from different positions (see Fig. 6). The virtual vehicle moved at constant speed, slower than the speed of the real car. The real vehicle accelerated up to 10 km/h and kept this speed constant as long as there was no obstacle in front it. The braking maneuver occurred when the real vehicle reached the minimum safety distance (2 meters) allowed to be compared to the car ahead. This distance value was transferred from the SUMO simulator to the controller via the CAN communication network. Practically, the CAN messages of the LIDAR and camera signals were overwritten by the traffic simulator's data. When the distance dropped below the safety distance the traffic jam assist logic stopped the real vehicle.

Obviously, the effectiveness of the test depends on the proper online transfer of speed and distance values. On the

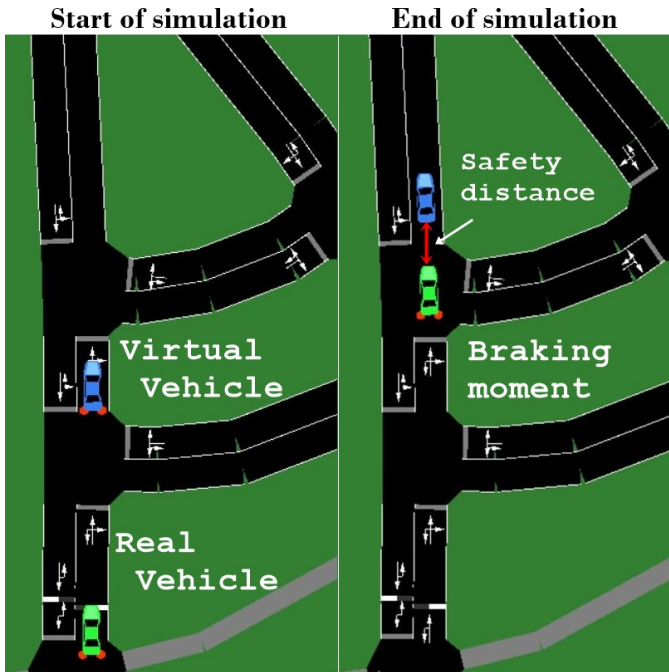


Fig. 6. The real world test scenario

one hand, the accuracy of the speed transmission provides the precise locomotion of the vehicle in the virtual reality. On the other hand, the distance value coming from the simulator means the input signal to the real car's controller. Hence, the success of the whole test is influenced by the reliable data transmission. For later evaluation all parameters sent and received during the simulation were logged. Fig. 7 shows the accuracy of the speed transfer. It can be seen that, as the result of communication at 5 Hz, a slight delay is present between the received CAN signal and the current speed value in SUMO. However, this delay is negligible, so that it did not affect the effectiveness of this test. Moreover, one can apply higher frequency of communication between the real car and the simulator (this is a question of computational resource). Fig. 8 shows the change of the distance between the two vehicles and the speed variation of the real autonomous test car. It is observable (according to the scenario) that after the start of the real vehicle the headway distance to the virtual car begins to decrease. Simultaneously the real car holds a constant speed until the safety distance limit is attained. Then the test car breaks down immediately.

The tools used during the experiment and the test site are shown by Fig. 9. Concerning the test vehicle, not all autonomous functions of the Smart were used. The self-driving controller of the test car handled the throttle and the break pedals as well as the gearbox, but steering was controlled manually. This is because the environment is not yet fully developed, i.e., implementation of autonomous turning manoeuvres is underway.

Simulations were performed on a common gamer laptop with Intel Core i5-6300HQ 2.3-GHz CPU (4 cores) and 12

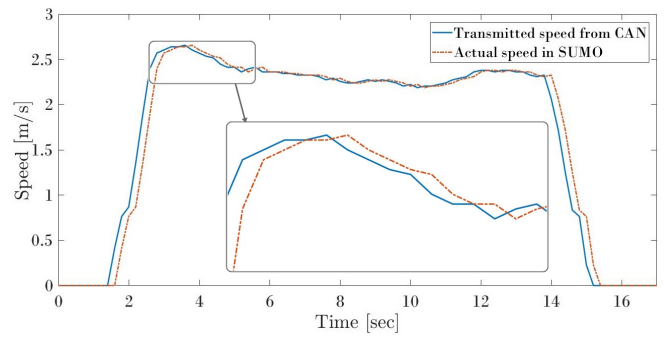


Fig. 7. The speed transmission between the vehicle and the simulator

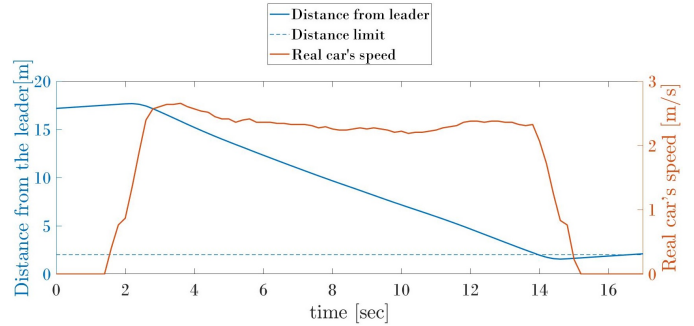


Fig. 8. The braking moment in the simulation

GB of RAM.



Fig. 9. The test environment during the experiment

V. CONCLUSION AND FUTURE WORK

The everyday use of autonomous cars must be preceded by broad manufacturer testing and validation. The research and development of autonomous vehicle technology are therefore important topics of our days and require extensive experimental works in the near future. By using the proposed VIL test environment, the research opportunities emerge by enabling easy configurable automotive tests. For further useful tests,

there are several options. Based on the CAN communication network, one can provide more information about the vehicle's environment, i.e., not only the position of virtual obstacles around the vehicle, but traffic management parameters (signal heads, road signs, etc.). It is also possible to model the communication of vehicles and infrastructure involved in traffic (e.g., V2V and V2I communication of the test vehicle). To achieve this, of course, there is a need for systems and controllers that can handle the signals generated by virtual reality. Another goal is to replicate the results now achieved with a 3-dimensional traffic simulator, real-time virtual imaging. This provides a better visual representation of the environment around the vehicle during VIL tests.

As a future work, the extension of the VIL test environment with GPS system is planned. This will allow GPS based speed and position determination as well (i.e., maps with GPS coordinates in the simulator). By maximizing the resources and increasing the frequency, a faster and more accurate communication can be created. Moreover, filtering techniques can also contribute to ameliorate the signal transmission between the elements.

REFERENCES

- [1] T. Tettamanti, I. Varga, and Zs. Szalay, "Impacts of autonomous cars from a traffic engineering perspective," *Periodica Polytechnica Transportation Engineering*, vol. 44, no. 4, pp. 244–250, 2016. [Online]. Available: <https://pp.bme.hu/tr/article/view/9464>
- [2] D. J. Fagnant and K. Kockelman, "Preparing a nation for autonomous vehicles: opportunities, barriers and policy recommendations," *Transportation Research Part A: Policy and Practice*, vol. 77, pp. 167 – 181, 2015. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0965856415000804>
- [3] A. Török, "Utility based decision making in autonomous vehicles," in *6th International Conference Towards a Human City: Smart Mobility - Synergy between Sustainable Mobility and New Technologies*, B. V. Bogdanović Vuk, Ed., no. 29-32. Novi Sad, Serbia: University of Novi Sad, Faculty of Technical Sciences, 25-27. October 2017.
- [4] Cs. Csiszár and D. Földes, "System model for autonomous road freight transportation," *PROMET - Traffic & Transportation*, vol. 30, no. 1, pp. 93–103, 2018.
- [5] M. M. Waldrop, "Autonomous vehicles: No drivers required," *Nature*, vol. 518, pp. 20–23, 2015.
- [6] P. Bansal and K. M. Kockelman, "Forecasting americans' long-term adoption of connected and autonomous vehicle technologies," *Transportation Research Part A: Policy and Practice*, vol. 95, pp. 49 – 63, 2017. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0965856415300628>
- [7] D. Milakis, M. Snelder, B. van Arem, B. van Wee, and G. de Almeida Correia, "Development of automated vehicles in the netherlands: scenarios for 2030 and 2050," *European Journal of Transport and Infrastructure Research*, vol. 17, no. 1, pp. 63–85, 2017.
- [8] C. Bartolini, T. Tettamanti, and I. Varga, "Critical features of autonomous road transport from the perspective of technological regulation and law," *Transportation Research Procedia*, vol. 27, pp. 791 – 798, 2017, 20th EURO Working Group on Transportation Meeting, EWGT 2017, 4-6. September 2017, Budapest, Hungary. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2352146517308992>
- [9] A. Nyerges and Zs. Szalay, "A new approach for the testing and validation of connected and automated vehicles," in *34th International Colloquium on Advanced Manufacturing and Repairing Technologies in Vehicle Industry*, no. 49-54, 17-19. May 2017, ISBN:978-963-313-258-6.
- [10] Zs. Szalay, T. Tettamanti, D. Esztergár-Kiss, I. Varga, and C. Bartolini, "Development of a test track for driverless cars: Vehicle design, track configuration, and liability considerations," *Periodica Polytechnica Transportation Engineering*, vol. 46, no. 1, pp. 29–35, 2018. [Online]. Available: <https://pp.bme.hu/tr/article/view/10753>
- [11] Zs. Szalay, A. Nyerges, Z. Hamar, and M. Hesz, "Technical specification methodology for an automotive proving ground dedicated to connected and automated vehicles," *Periodica Polytechnica Transportation Engineering*, vol. 45, no. 3, pp. 168–174, 2017. [Online]. Available: <https://pp.bme.hu/tr/article/view/10708>
- [12] *Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles*, SAE International Std. J3016, 09 2016. [Online]. Available: <https://www.sae.org/standards>
- [13] R. J. F. R. José L. F. Pereira, "An integrated architecture for autonomous vehicles simulation," *Proceedings of the 27th annual ACM symposium on applied computing*, pp. 286–292, 2014.
- [14] U. N. Pedro Fernandes, "Platooning of autonomous vehicles with inter-vehicle communications in sumo traffic simulator," *Annual Conference on Intelligent Transportation Systems*, 2010.
- [15] Zs. Szalay, "Structure and architecture problems of autonomous road vehicle testing and validation," in *15th Mini Conference on Vehicle System Dynamics, Identification and Anomalies (VSDIA)*, Hungary, Budapest, 7-9. November 2016.
- [16] D. Krajzewicz, J. Erdmann, M. Behrisch, and L. Bieker, "Recent development and applications of sumo - simulation of urban mobility," *International Journal On Advances in Systems and Measurements*, vol. 5, no. 3-4, pp. 128–138, December 2012. [Online]. Available: <http://elib.dlr.de/80483/>
- [17] A. Wegener, M. Piórkowski, M. Raya, H. Hellbrück, S. Fischer, and J.-P. Hubaux, "Traci: An interface for coupling road traffic and network simulators," in *Proceedings of the 11th Communications and Networking Simulation Symposium*, ser. CNS '08. New York, NY, USA: ACM, 2008, pp. 155–163. [Online]. Available: <http://doi.acm.org/10.1145/1400713.1400740>
- [18] V. Tihanyi and Zs. Szalay, "Autonomous vehicle platform for demonstration purposes," in *34th International Colloquium on Advanced Manufacturing and Repairing Technologies in Vehicle Industry*, Visegrád, Hungary, May 2017, pp. 145–148.